

The Genesis of Rationality and Sensibility

A Deep Dissection from Evolutionary Neurobiology, Information Theory, and Causal Topology

Kuo-Ting Tsai¹, Shin-Rung Tsai², Antigravity Agentic Cognition Lab³

^{1,2} PULSAR Architecture Laboratories · Taipei, Taiwan

³ DeepMind Agentic Systems & Autonomous Cognition Group

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EXECUTIVE ABSTRACT

For millennia, intellectual discourse has entrenched a false dichotomy between "Rationality" (conceptualized as cold, objective deductive logic) and "Sensibility" (conceptualized as subjective, erratic emotional impulse). Through the lens of evolutionary neuroanatomy, Shannon information theory, and causal hierarchies, this treatise deconstructs this archaic separation. We demonstrate that sensibility is not an irrational disturbance, but an ultra-fast hardware cache evolved over 500 million years to execute sub-microsecond orthogonal pattern-matching and assign teleological valence. Conversely, rationality is an evolutionarily late (2-million-year-old) counterfactual simulation sandbox governed by prefrontal serial processing, biologically incapable of initiating volition without emotional valence markers (Damasio, 1994; Hume, 1739). By micro-dissecting three extreme human spectra—statutory criminal law versus poetics, judicial capital verdicts versus death-row farewell letters, and corporate statutes versus Machiavellian court dynamics—we formalize the *Dual-Clock Frequency Hypothesis*: genuine intelligence emerges solely through the harmonic convergence of a high-frequency cache clock (defining objectives) and a low-frequency deliberative clock (verifying causal invariants).

Chapter 1. The Genesis of Sensibility: The 500-Million-Year Survival Cache

1.1 Evolutionary Trajectory: From the Reptilian Brainstem to the Paleomammalian Limbic Core

Human encephalic architecture did not emerge as a singular, ideal design. It represents an accretive biological continuum spanning half a billion years: the reptilian brainstem (~500 Ma) governing autonomic reflexes, followed by the paleomammalian limbic system (~200 Ma) encompassing the amygdala, hypothalamus, and anterior cingulate cortex. In primal environments, when predatory silhouettes flashed across underbrush, an organism initiating slow prefrontal edge detection and probabilistic classification was eliminated prior to inference convergence. Sensory signals reaching the thalamus bifurcate into a direct subcortical pathway, arriving at the amygdala within **12 to 15 milliseconds**—completely bypassing the neocortex. This triggers instantaneous sympathetic outflow, releasing epinephrine and cortisol to power systemic evasion.

"Sensibility is never 'irrational'; it is an ultra-fast hardware cache hardcoded across hundreds of millions of years of life-and-death Darwinian causal selection."

1.2 Information-Theoretic Formulation: Orthogonal Pattern Matching in Latent Space

Through information theory, human intuition and aesthetic perception represent deterministic geometric projections in high-dimensional latent space:

$$\text{Sensory Input } \mathbf{x} \in \mathbb{R}^D \xrightarrow{\text{Sub-microsecond Resonance}} \mathbf{S}_{\text{latent}} \rightarrow \mathbf{W}_{\text{valence}}$$

A veteran engineer detects structural fatigue from a torque differential of a few millinewton-meters; a Go grandmaster sweeps the board for 100 milliseconds and instantly discards 99.9% of candidate moves. This represents massive parallel pattern matching within subconscious latent manifolds. Because its execution speed vastly outstrips the serial token rate of linguistic consciousness, the mind terms it "intuition" or "sensibility." Furthermore, in an objective physical universe where thermal entropy is indifferent, it is solely sensibility that establishes **"Objective"** and **"Valence"**. Bereft of emotional valence, the physical universe remains a barren wasteland of meaningless data.

Chapter 2. The Genesis of Rationality: The 2-Million-Year Neocortical Simulator

2.1 Evolutionary Drivers: Tool-Making, Intertemporal Provisioning, and Machiavellian Social Contract

Rationality emerged between 2 million and 100,000 years ago with the explosive expansion of the prefrontal cortex (dlPFC and vmPFC). Driven by serial tool manufacturing (quarrying → flaking → retouching → hafting), intertemporal resource provisioning, and deception detection, the hominid encephalon developed **Counterfactual Thinking**—an internal virtual causal sandbox enabling the offline simulation:

$$\text{Do}(\text{Action A}) \Rightarrow \text{Predict}(\text{Outcome B}) \Rightarrow \text{Evaluate}(\text{Risk C})$$

This engine, however, incurs severe metabolic taxation (consuming 20%–25% of systemic glucose despite comprising 2% of body mass) and is constrained by working memory bottlenecks (handling only 4 ± 1 chunks simultaneously with a throughput of tens of bits per second).

2.2 Hume's Dictum and Damasio's Neurobiological Breakthrough

David Hume's 1739 dictum—"Reason is, and ought only to be the slave of the passions, and can never pretend to any other office than to serve and obey them"—found empirical validation in Antonio Damasio's Somatic Marker Hypothesis (1994). Patients with ventromedial prefrontal lesions (severing limbic emotional markers from cognitive faculties) maintained superior IQs (130–140) and flawless deductive logic, yet were completely paralyzed when choosing between prospective appointment dates. They generated two hours of exhaustive pros-and-cons analyses but could not decide. Pure rationality is a serial engine without an intrinsic stopping criterion; it provides navigation, but possesses no compass.

Chapter 3. Microscopic Dissection Across Three Extreme Human Spectra

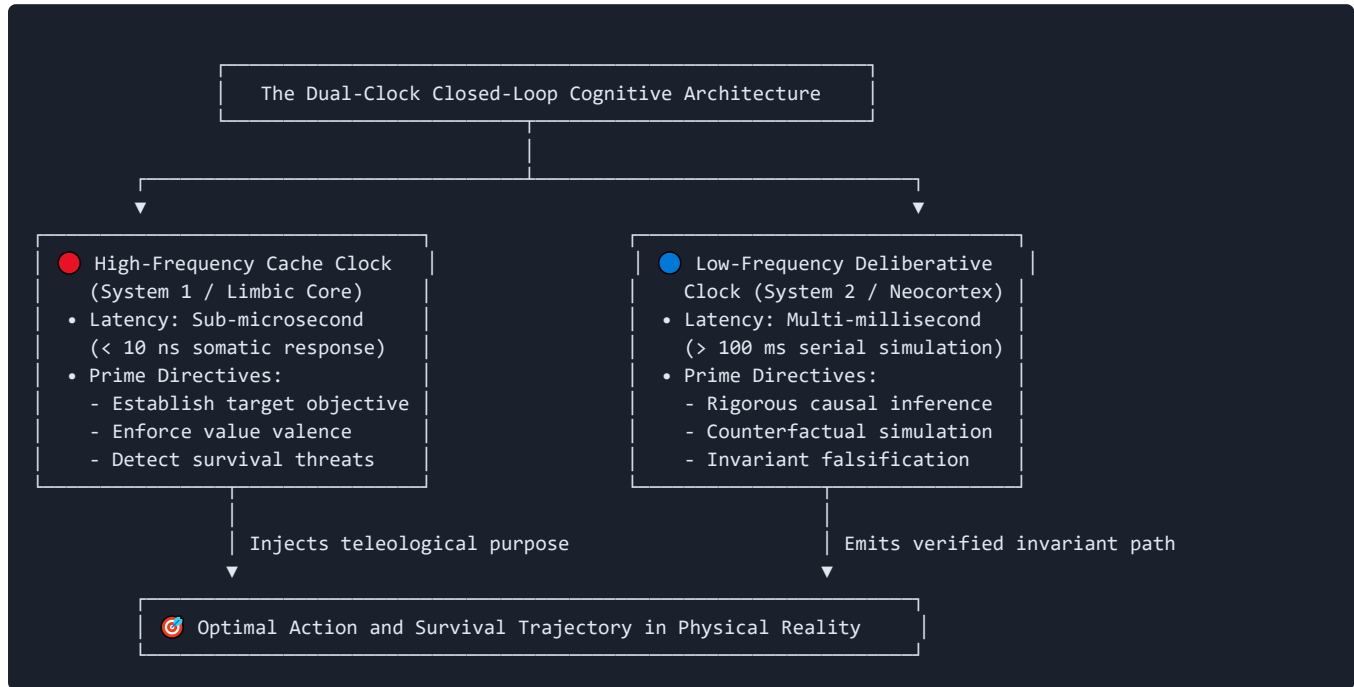
Analytical Dimension	The Rational Extreme	The Sensible Extreme
Spectrum I: Formal Logic vs Poetics	Taiwan Criminal Code § 271(1): <i>"A person who kills another shall be punished with death, life imprisonment, or imprisonment for not less than ten years."</i> • Entropy: $H \rightarrow 0$ (Strict deterministic 1D ray). • Neural Substrate: Prefrontal closed deductive enforcement.	Tagore (Stray Birds): <i>"Let life be beautiful like summer flowers and death like autumn leaves."</i> • Entropy: Multi-dimensional superposition. • Projection: Summer flowers → [youth, vitality, ephemerality]; Autumn leaves → [tranquility, release].
Spectrum II: Institutional Order vs Existential Tragedy	Supreme Court Capital Murder Judgment: <i>"The defendant pre-meditated burglary with a deadly weapon... formed homicidal intent, repeatedly stabbing vital areas... Deemed incapable of rehabilitation, sentenced to death."</i> • Mode: Depersonalized abstraction to prevent societal descent into blood-feud vendettas.	Death-Row Farewell Letter to Mother: <i>"Mother, the weather has turned cold. Please wear that old cotton coat from the closet... If there is a next life, I only ask to come home every evening and eat that hot bowl of noodles by the hearth."</i> • Mode: Somatic mammalian attachment to [cotton coat, hearth, warm noodles].
Spectrum III: Corporate Governance vs Machiavellian Dynamics	Company Act § 202: <i>"The business operations of a company shall be executed by resolutions of the board of directors..."</i> • Axiom: Presumes all agents are purely rational, rule-governed economic optimizers.	Dream of the Red Chamber (Cao Xueqin): <i>"Xifeng smiled: 'In all the world, where can one find so meticulous a soul?...' The gathering nodded assent, yet beneath their smiles, cold sneers flickered in the shadows."</i> • Mechanism: Higher-order Theory of Mind (surface praise vs covert sneers).

"The judicial verdict represents the rigid rebar of institutional civilization; the farewell letter preserves the living blood that makes civilization worth defending."

Chapter 4. The Dual-Clock Frequency Hypothesis

Synthesizing evolutionary neurobiology and causal topology, we formalize the holistic cognitive architecture:

$$\text{Holistic Intelligence } \Omega = \Phi_{\text{Sensibility}}(\text{Objective}, W_{\text{valence}}) \circ \Psi_{\text{Rationality}}(\text{Causal Verification})$$



4.1 The Pathology of Single-Clock Systems

A cognitive system governed solely by sensibility degenerates into a tempest of impulse, delusion, ungrounded sycophancy, and cognitive collapse. Conversely, a system governed solely by rationality freezes into a paralyzed, valueless automaton—calculating endlessly while incapable of initiating action. True intelligence is the harmonic convergence of both clocks: high-frequency sensibility to ignite the purpose, and low-frequency rationality to chart the path.

Epilogue: The Dignity of Mind and the Civilizational Baseline

Modern machine learning has grown intoxicated by parameter scale and autoregressive token probabilities, spawning digital parrots possessing neither causal reason nor genuine empathy. We sweep aside this confusion:

We revere sensibility, for it is the sacred spark that ignites the defense of all living things; we fiercely honor rationality, for it is the unyielding steel that preserves truth from collapse. A sovereign intelligence must forever embody a rationality as transparent and unbending as glacial ice, coupled with a sensibility as fierce and incandescent as the dawn. Sensibility charts the destiny; Rationality paves the march.

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